Supplementary Material

Downstream Utility of Synthetic Data for Clustering

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Abstract

Supporting figures and tables for the main article.

S1 Recovery of the number of groups by cluster count and separation

Figure S1 presents the recovery rate of the true number of groups as a function of the true cluster count k and the separation σ, for both algorithms. It complements panel (a) of Figure 3 of the main article in a different layout, making the $k \times \sigma$ interaction directly legible: recovery collapses in the lower-left quadrant (many groups, little separation) and is essentially perfect in the upper-right.

S2 Agreement in the estimated number of groups between original and synthetic data

Table S1 cross-tabulates the number of groups estimated from the original data against the number estimated from the corresponding synthetic data, stratified by separation.

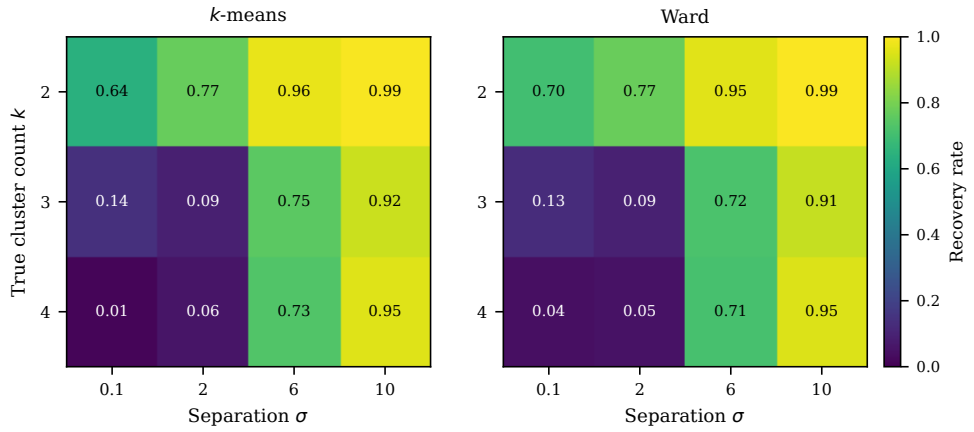


Fig. S1 Recovery rate of the true number of groups by true cluster count and separation, for k-means (left) and Ward (right). Cell values are proportions over the full design.

Unlike the recovery rate in the main article, which scores $\hat{k}$ against the construction-time ground truth, this table compares the two estimates with each other and so isolates the effect of synthesis from the difficulty of the clustering problem itself.

The pattern confirms the expectation that at high separation the mass concentrates on the diagonal: exact agreement rises from 68.4% at $\sigma = 0.1$ to 98.2% at $\sigma = 10$ for k-means (Ward: 64.6% to 97.5%). In the overlapping regime the off-diagonal mass is not symmetric. At $\sigma = 0.1$ the synthetic data yield a *lower* estimate than the original in 15.2% of comparisons and a higher one in 16.4%; the largest single off-diagonal cell is the original data supporting $\hat{k} \geq 5$ while the synthetic data support $\hat{k} = 2$ (8.4%). That is, when the original data contain weak, diffuse structure that the silhouette criterion resolves into many small groups, CART synthesis tends to smooth it away into a single coarse split. The tendency disappears once the groups are genuinely separated.

S3 Sensitivity to the candidate range for the number of groups

In the main article the silhouette criterion searches $\hat{k} \in \{2, \ldots, \min(k + 3, 8)\}$, a rule inherited from earlier work that ties the search range to the planted number of groups. Table S2 repeats the count analyses with the fixed range $\{2, \ldots, 8\}$, applied identically to original and synthetic data. Widening the range lowers both agreement measures in the overlapping regime, where the silhouette criterion occasionally selects large values of $\hat{k}$, and leaves them essentially unchanged at $\sigma \geq 6$. The qualitative conclusions are unaffected: the two algorithms remain indistinguishable against the planted number of groups and k-means remains significantly more stable against the original data.

S4 Independent-resampling baseline

To judge whether a given level of agreement is high, it helps to know how often two datasets that differ only by sampling variability yield the same estimate. Within each scenario we compare the estimates $\hat{k}_R$ from the $n = 5$ independent original datasets and record the proportion of the 10 pairs that coincide. Table S3 sets this baseline beside the synthetic-versus-original agreement AG. Outside the most overlapping regime, a CART synthetic dataset agrees with the original dataset from which it was generated more often than two independent samples from the same population agree with each other. At $\sigma = 0.1$ the two are comparable for Ward, and synthetic agreement is lower for k-means. The baseline rests on only 10 pairs per scenario and is therefore a coarse reference rather than a formal null distribution.

Table S1 Agreement in the estimated number of groups between original and synthetic data, by separation. Rows are $\hat{k}$ from the original data, columns $\hat{k}$ from the synthetic data; cells are percentages of all comparisons at that separation (each block sums to 100%). Estimates of five or more groups are pooled for display, but the agreement column is computed on unpooled values. Both data sources are searched over a fixed range $\hat{k} \in \{2, \ldots, 8\}$, so the comparison is symmetric; under the narrower per-scenario range used for the recovery rates in the main text, agreement is higher in the overlapping regime (75.1% and 67.6% at $\sigma = 0.1$, 79.7% and 75.9% at $\sigma = 2$, for k-means and Ward respectively) and essentially unchanged at $\sigma \geq 6$, where the two conventions differ by at most 0.1 percentage points. $n = 180{,}000$ comparisons per separation level.

		$\hat{k}$ recovered from synthetic data (%)				Exact
σ	$\hat{k}$ (original)	2	3	4	≥ 5	agreement
k-means						
0.1	2	50.9	2.1	0.5	3.8	68.4%
	3	2.2	10.0	0.5	4.0	
	4	0.0	0.0	0.0	0.0	
	≥ 5	8.4	1.1	0.6	16.0	
2	2	62.9	3.4	1.6	4.2	76.4%
	3	1.2	8.6	0.7	2.2	
	4	0.3	0.4	0.9	1.2	
	≥ 5	2.5	0.9	0.9	8.0	
6	2	38.1	0.9	0.3	0.1	92.6%
	3	0.7	29.1	1.1	0.1	
	4	0.5	2.4	24.7	0.8	
	≥ 5	0.0	0.1	0.2	0.8	
10	2	34.8	0.2	0.2	0.4	98.2%
	3	0.2	31.7	0.3	0.0	
	4	0.0	0.2	31.7	0.3	
	≥ 5	0.0	0.0	0.0	0.0	
Ward						
0.1	2	58.3	5.0	2.0	6.9	64.6%
	3	3.6	4.6	1.1	3.5	
	4	1.2	0.9	0.4	0.9	
	≥ 5	6.3	1.3	0.6	3.4	
2	2	65.7	4.7	2.1	5.3	73.8%
	3	1.1	6.4	1.0	2.1	
	4	1.1	0.5	0.4	1.4	
	≥ 5	2.6	1.0	0.8	3.9	
6	2	37.1	1.0	0.2	0.1	88.2%
	3	1.9	27.3	1.6	0.3	
	4	1.8	2.7	23.6	1.3	
	≥ 5	0.3	0.5	0.1	0.3	
10	2	34.8	0.6	0.1	0.0	97.5%
	3	0.3	31.1	0.3	0.0	
	4	0.0	0.3	31.6	0.4	
	≥ 5	0.4	0.0	0.0	0.2	

Table S2 Recovery of the planted number of groups (RR) and agreement with the number estimated from the original data (AG), by separation, under the per-scenario candidate range used in the main text and under a fixed range. Wilcoxon p: paired signed-rank test of k-means versus Ward across the 144 scenario-level means.

	$\hat{k} \in \{2, \ldots, \min(k+3, 8)\}$				$\hat{k} \in \{2, \ldots, 8\}$			
	RR		AG		RR		AG	
Separation	k-m.	Ward	k-m.	Ward	k-m.	Ward	k-m.	Ward
$\sigma = 0.1$	0.272	0.296	0.751	0.676	0.247	0.281	0.684	0.646
$\sigma = 2$	0.304	0.307	0.797	0.759	0.288	0.295	0.764	0.738
$\sigma = 6$	0.810	0.789	0.926	0.882	0.810	0.789	0.926	0.882
$\sigma = 10$	0.954	0.950	0.983	0.976	0.954	0.950	0.982	0.975
Overall	0.585	0.585	0.864	0.823	0.575	0.579	0.839	0.811
Wilcoxon p	0.39		< 0.001		0.76		< 0.001	

Table S3 Synthetic-versus-original agreement (AG) and the independent-resampling baseline (proportion of pairs of independent original datasets in the same scenario with equal $\hat{k}_R$). Means over scenarios; candidate range $\{2, \ldots, \min(k+3, 8)\}$.

		k-means		Ward	
Stratum	Level	AG	Baseline	AG	Baseline
Separation σ	0.1	0.751	0.817	0.676	0.681
	2	0.797	0.750	0.759	0.728
	6	0.926	0.764	0.882	0.758
	10	0.983	0.933	0.976	0.922
Distribution	Normal	0.896	0.872	0.898	0.869
	Gamma	0.833	0.760	0.749	0.675
Overall		0.864	0.816	0.823	0.772